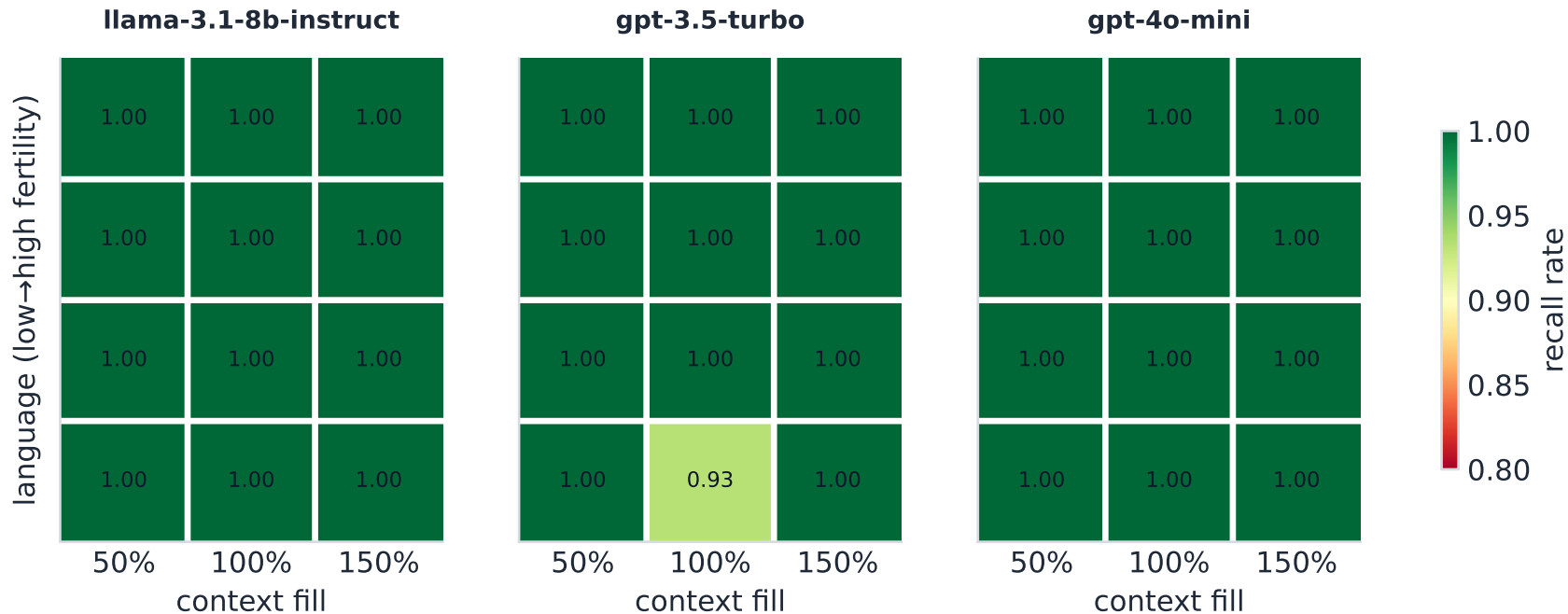


Figure 4. Measured needle recall — no degradation at ≤ 12 k tokens



Mean recall over needle positions 5–95%; base window 8192 (fill 50/100/150% $\approx$ 4k/8k/12k tokens). Every model ≥ 0.99 across all languages; the lone dip (gpt-3.5 / Malayalam, 0.93) is a single trial. Consistent with FertiScope's window-dependent risk model: within the models' windows the fertility tax does not yet cause recall loss — degradation is a near/beyond-window phenomenon (future work).